

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2024-2025
FULL MARKS: 150

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) What are the steps of problem-solving in engineering? With a suitable example, briefly explain how numerical methods are used to solve an engineering problem. 3 + 2
(CO2)
(PO1)
- b) What are the drawbacks of the Bisection method? Explain with proper examples and illustrations. 6
(CO4)
(PO1)
- c) Consider two ladders of known lengths $a > 0$ and $b > 0$ leaning across a straight alley of unknown width $x > 0$, touching opposite vertical walls, as portrayed in Figure 1. The triangle bounded by the ground and the lower sections of the two ladders has a known area $A > 0$. Assume the configuration is physically realizable and that $0 < x < \min(a, b)$.

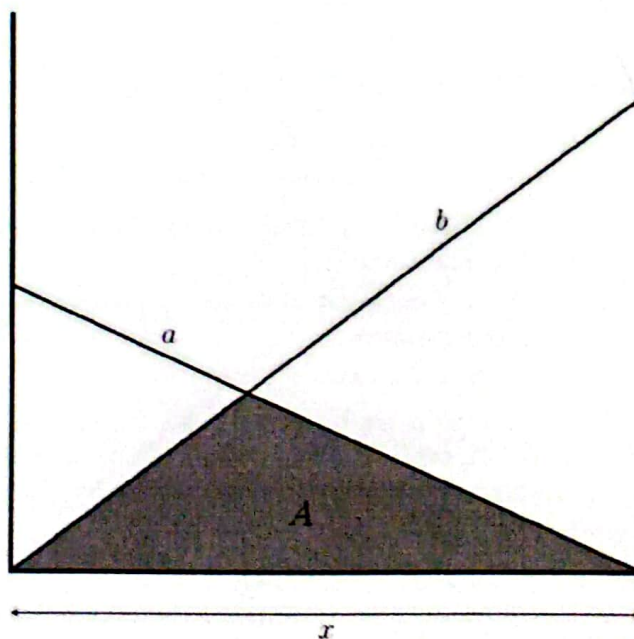


Figure 1: The crossed ladders for Question 1.c

- i. From the given scenario, formulate a function $f(x)$ (in terms of a , b , and A) whose positive root yields the width x of the alley. 12
(CO1)
(PO1)
- ii. Derive the Newton-Raphson formula for iteratively solving $f(x) = 0$, using the function f formulated in the preceding Question 1.c)i. 6
(CO2)
(PO1)

2. a) Define Round-off Error. What are the different ways through which Round-off Error can be introduced? Explain with proper examples. 5
(CO4)
(PO1)

b) The Maclaurin series for $\sin(x)$ is given as follows:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

- i. Why is the sinusoid function $\sin(x)$ categorized as a transcendental function? 2
(CO4)
(PO1)
- ii. Use the Remainder Theorem to find bounds on the truncation error in the representation of $\sin(2)$ if only the first 3 terms of the Maclaurin series are used. 5
(CO2)
(PO1)



The Taylor series for a function $f(x)$ is —

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \operatorname{cosec}(x)$$

3. a) Prove that a polynomial of degree n or less that passes through $n+1$ data points is unique. 6
(CO4)
(PO1)

- b) i. What is the Runge phenomenon? 2
(CO3)
(PO1)

(ii) A natural cubic spline $S_3(x)$ on $[0, 2]$ is defined by:

$$S_3(x) = \begin{cases} 1 + 2x - x^3; & \text{if } 0 \leq x \leq 1, \\ 2 + a(x-1) + b(x-1)^2 + c(x-1)^3; & \text{if } 1 \leq x \leq 2. \end{cases}$$

Given that $S_3(x)$ is a natural cubic spline, it must satisfy the following conditions:

- $S_3(x)$ is continuous at the interior point $x = 1$,
- The second derivative $S_3''(x)$ vanishes at the exterior points $x = 0$ and $x = 2$, as it exhibits no curvature at those points.

Find the values of the constants a , b , and c .

- c) A robot arm with a rapid laser scanner is performing a quick quality check on holes drilled in a $15'' \times 10''$ rectangular plate. The centers of the holes describe the path the arm needs to follow. The hole centers are located on a Cartesian coordinate system (with the origin at the bottom left corner of the plate) as given in Table 1. 8
(CO2)
(PO1)

Table 1: Hole coordinates for Question 3.c

Abscissa, x (in)	2	4.25	5.25	7.81	9.2	10.6
Ordinate, y (in)	7.2	7.1	6	5	3.5	5

If the laser is traversing from $x = 2$ to $x = 4.25$ to $x = 5.25$ along a quadratic path, determine the value of y at $x = 4$ using the Newton's Divided Difference method of interpolation with a second order polynomial.

4. a) The general linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is 7+5
(CO4)
(PO1)

$$y = a_0 + a_1x$$

where a_0 and a_1 are the two parameters of the regression model.

- i. Derive the formulae for finding the optimal values of a_0 and a_1 .
 - ii. Prove that the values of a_0 and a_1 obtained using the formulae derived in Question 4.a)i correspond to the absolute minimum of the optimization criterion used.
- b) In the case of the Logistic Growth model, determine the values of y that correspond to $x = 0$ and $x \rightarrow \infty$. 5
(CO3)
(PO1)
- c) Suppose, you are given n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ and you want to regress them to the Exponential model $y = e^{bx}$. 5
(CO3)
(PO2)
- Without performing any data transformation, you obtain the regression model $y = e^{k_1x}$.
 - With data transformation via linearization, you obtain the regression model $y = e^{k_2x}$.
- With proper reasoning, explain whether k_1 and k_2 are equal or not.
5. a) i. Using the method of coefficients, derive the formula for the Single-segment Trapezoidal rule. 7
(CO2)
(PO1)
- ii. Applying the formula from Question 5.a)i, derive the formula for the Multi-segment Trapezoidal rule. 6
(CO2)
(PO1)

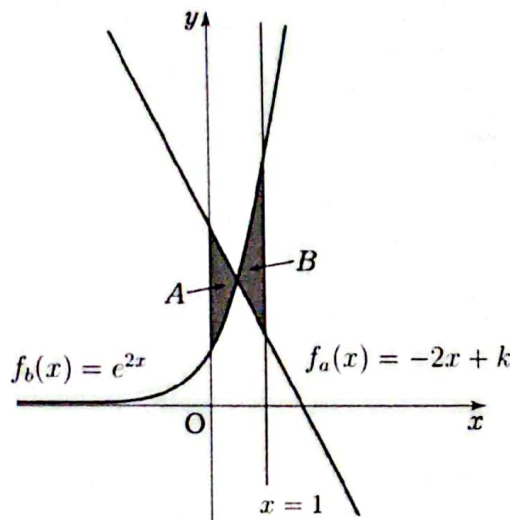


Figure 2: The shaded regions A and B for Question 5.b.

- Ⓒ In Figure 2, the area of the shaded regions A and B, portrayed beneath the two functions $f_a(x)$ and $f_b(x)$ respectively, are equal. The aforementioned functions are, 8 + 6
(CO2)
(PO1)
- $$f_a(x) = -2x + k$$
- $$f_b(x) = e^{2x}$$
- i. Using the Composite Simpson's 3/8 Rule, with $n = 9$ segments, approximate the value of k .
 - ii. Analytically determine the exact value of k , denoted by k_{exact} , and use it to calculate the true error (E_t) and the absolute relative true error ($|e_t|\%$).

6. a) A ball at 1500 K is allowed to cool down in air at an ambient temperature of 500 K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by 8
(CO2)
(PO1)

$$\frac{d\theta}{dt} = -1.2 \times 10^{-12} (\theta^4 - 625 \times 10^8)$$

where θ is in K and t in seconds. Find the temperature at $t = 200$ seconds using Runge-Kutta 2nd order method (Heun's version). Assume a step size of $h = 100$ seconds.

(b)

- i. What is the LU decomposition of A, where 6
(CO2)
(PO1)

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

- ii. Determine the order and degree of each of these higher-order differential equations 0.5 × 4

(a) $x^2 \frac{dy}{dx} - \left(\frac{d^3y}{dx^3} \right)^3 + 69 = 0$

(b) $\left(\frac{dy}{dx} \right)^2 - 420xy = 3$ (CO4)
(PO1)

(c) $\frac{d^3y}{dx^3} + x^4 \frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^4 = \sin x$

(d) $\frac{dy}{dx} + xy = x^2$

- (c) Derive the orders of error (in Big O notation) for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for determining the best one of the three aforementioned methods. 6 + 2
(CO2)
(PO1)

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Math 4641: Numerical Methods

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 Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) Suppose you are given the data shown in Table 1:

Table 1: t vs $f(t)$ data for Question 1.a)

t	$f(t)$
0	0
0.5	0.19
1.0	0.26
1.5	0.29
2.0	0.31

- i. Using Taylor series, derive the Central Difference Approximation formula for $f''(x_i)$. 7
(CO1)
(PO1)
 2×3
(CO2)
(PO1)
- ii. Find $f'(1.5)$ and $f''(1.5)$. 7 + 5
(CO3)
(PO1)
- b) Prove whether Central Difference Approximation produces less error than Forward Difference or Backward Difference Approximation. Explain why Forward Difference and Backward Difference formulae may still be needed even though Central Difference formula produces less error than both. 5
(CO1)
(PO1)
2. a) i. Rewrite the following third-order differential equation as a set of simultaneous first-order differential equations: 14
(CO2)
(PO1)
- $$\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = e^{-t}$$
- $$y(0) = 1$$
- $$\frac{dy}{dx}(0) = 2$$
- $$\frac{d^2y}{dx^2}(0) = 2$$
- ii. For the differential equation in Question 2.a)i, find $y(0.75)$ using Euler's method. Consider step size, $h = 0.25$. 2×3
(CO1)
(PO1)
- b) Show how the variables are transformed when linearization is applied on Exponential function, Power function and Growth function. 2×3
(CO1)
(PO1)

3. a) Explain how LU Decomposition is more computationally efficient for solving linear systems compared to Inverse matrix multiplication or Gaussian Elimination.

7
(CO3)
(PO1)

- b) Suppose you are given the time vs velocity data for two separate missiles in Table 2.

18
(CO2)
(PO1)

Table 2: Time vs Velocity data for two missiles for Question 3b)

t	V_1	V_2
0	0	0
10	227.04	152.35
15	362.78	316.62
20	517.35	502.71
22.5	602.97	620.25
30	901.67	978.87

Using Direct method of Interpolation with a third order polynomial, find the velocity of both missiles at $t = 21$ seconds. For solving the simultaneous equations, you are not allowed to use inverse matrix multiplication or Gaussian Elimination more than once.

4. a) i. What is the geometrical interpretation of Eigenvalues and Eigenvectors for a given matrix? Explain with an example.

6
(CO1)
(PO1)

- ii. Provide mathematical reasoning as to whether $\lambda = 3$ is one of the four eigenvalues of the following matrix A:

12
(CO2)
(PO1)

$$A = \begin{bmatrix} 4 & 3 & 6 & 3 \\ 3 & 4 & 6 & 2 \\ 4 & 2 & 3 & 2 \\ 2 & 3 & 6 & 5 \end{bmatrix}$$

- b) Find the eigenvalues and eigenvectors of the following matrix B:

$$B = \begin{bmatrix} 6 & -3 \\ 1 & 4 \end{bmatrix}$$

7
(CO2)
(PO1)

5. a) Find the following integral using Simpson's 3/8 Rule:

$$I = \int_0^{100} f(x) dx$$

13
(CO2)
(PO1)

where,

$$f(x) = \begin{cases} 0, & 0 < x < 30 \\ -9.1688 \times 10^{-6}x^3 + 2.796 \times 10^{-3}x^2 - 0.28487x + 9.6778, & 30 < x < 172 \end{cases}$$

- b) Solve the integral given in Question 5.a) using Euler's method considering step size, $h = 25$.

12
(CO2)
(PO1)

$$y_{i+1} = y_i + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \Delta t$$

6. a) When a voltage input of 20V is provided, the speed of a motor is obtained by the following differential equation: 12

$$20 = 0.02 \frac{dv}{dt} + 0.06v$$

(CO2)
(PO1)

If the initial speed of the motor was 0, then find the speed at $t = 0.8$ using Runge-Kutta 4th order method taking step size, $h = 0.4$.

- b) For drawing road networks from aerial images, light intensities are measured at different pixel locations. The following intensities are given as a function of pixel location in Table 3: 13
(CO2)
(PO1)

Table 3: Pixel location vs Intensity data for Question 6.b)

Pixel (k)	Intensity (y)
-3	119
-2	165
-1	231
0	243
1	244
2	214
3	136

Suppose we want to perform polynomial regression on the given data using a 2nd order polynomial:

$$y = a_0 + a_1k + a_2k^2$$

Find the corresponding polynomial model parameters.

key

$$y = a_0 + a_1k + a_2k^2$$

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Answer all **6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin.

1. a) A Trunnion of diameter 12.363" has to be cooled from a room temperature of 80°F before it can be shrink-fit into a steel hub. The equation that gives the diametric contraction (in inches) of the trunnion in dry-ice (boiling point -108°F) is: 8+10
(CO2)
(PO1)

$$\Delta D = 12.363 \int_{80}^{-108} (-1.2278 \times 10^{-11} T^2 + 6.194 \times 10^{-9} T + 6.015 \times 10^{-6}) dT$$

Here, a positive ΔD indicates expansion and negative value indicates contraction.

Find ΔD using the 2-segment and 4-segment Simpson's 1/3 rule.

- b) Find the relative approximate error between the result of 2-segment and 4-segment Simpson's 1/3 rule. Analyze the reason for getting such low error. 4+3
(CO3)
(PO2)
2. a) To simplify a model for a diode, it is approximated by a forward bias model consisting of a DC voltage, V_d , and a resistor, R_d . Table 1 contains the voltage vs current data that is collected for a small signal: 12+4
(CO2)
(PO1)

Table 1: Voltage vs Current

V (volts)	I (amps)
0.6	0.01
0.7	0.05
0.8	0.2
0.9	0.7
1.0	2.0
1.1	4.0

Current can be linearly regressed from voltage data as $I = B_0 + B_1 V$, where B_0 and B_1 are the constants of regression. Once B_0 and B_1 are known, the voltage and resistance for the diode can be computed as $V_d = -\frac{B_0}{B_1}$ and $R_d = \frac{1}{B_1}$.

Find the voltage and resistance for the diode.

- b) Explain how relative error can provide more useful insight compared to true error when analyzing the results of a numerical method. 9
(CO3)
(PO2)
3. a) Derive the Trapezoidal rule of Integration from both Calculus and Method of Coefficients. Analyze the geometrical interpretation of these two derivations. 10+5
(CO1)
(PO2)
- b) Taking upto 5 terms of the Taylor series, find the value of $\cos(\pi)$ given that we know $\cos\left(\frac{\pi}{2}\right) = 0$. 10
(CO2)
(PO1)
4. a) Thermistors are used to measure the temperature of bodies. These instruments are based on materials' change in resistance with temperature. To measure temperature, manufacturers provide you with a temperature vs. resistance calibration curve. If you measure resistance, you can find the temperature. Several observations with a thermistor are shown in Table 2. 11
(CO2)
(PO1)

Table 2: Resistance vs Temperature

R (ohm)	T (°C)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the temperature for 754.8 ohms using a 1st order Lagrange polynomial.

- b) Using the previous data, determine the result for the same resistance (754.8 ohms) using a 2nd order Lagrange polynomial and find the absolute relative approximate error.

11+3
(CO3)
(PO2)

5. a) What is the minimum number of points needed to evaluate the integral for Simpson's 3/8 rule? Derive the Multiple Segment Simpson's 3/8 rule.

3+10
(CO2)
(PO1)

- b) Suppose you are given the following data:

Table 3: x vs F(x)

x	0	0.5	1.0	1.5	2.0
F(x)	0	0.19	0.26	0.29	0.31

3×4
(CO2)
(PO1)

Find the Forward Difference, Backward Difference and Central Difference approximations of $F'(1)$ using $h = 0.5$.

6. Suppose you are given the following matrix A:

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

- a) Now, answer the following questions:

- i) Analyze the properties of the matrix A to determine whether it can have complex eigenvalues.
- ii) Find the eigenvalues and eigenvectors of the matrix A.

4
(CO1)
(PO2)
12
(CO2)
(PO1)

- b) Find the LU Decomposition of the matrix A.

9
(CO2)
(PO1)

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1. a) Use Taylor's polynomial to derive the orders of error for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for choosing one of these three. 9+5
(CO1, CO3)
(PO1)
- b) A tissue paper company advertises that every roll of their toilet paper has at least 250 sheets. The probability that there are 250 or more sheets in the toilet paper roll is given by the following equation: 11
(CO2)
(PO2)

$$P(y \geq 250) = \int_{250}^{\infty} 0.3515e^{-0.3881(y-252.2)^2} dy$$

By theoretical reasoning, the above integral calculates the probability up to infinite sheets. But practically, it has been seen that the number of sheets rarely exceed 270. So, considering 270 as the maximum possible number of sheets, the above integral can be approximated as follows:

$$P(y \geq 250) = \int_{250}^{270} 0.3515e^{-0.3881(y-252.2)^2} dy$$

Use the 2-segment Trapezoidal rule to find the probability that there are 250 or more sheets in a random toilet paper roll.

2. a) Without using linearization transform, describe the individual solution approaches for the following nonlinear regression models: 4+5+4
(CO1)
(PO1)
 - i. Exponential model
 - ii. Growth model
 - iii. Polynomial model
- b) What is the drawback in directly using the first 3 terms of the Taylor series for deriving y_{i+1} in Runge-Kutta 2nd order method? What is the alternative way of approximating the first 3 terms of the series? 3+3
(CO3)
(PO2)
- c) What are the different forms of the approximation of Runge-Kutta 2nd order method based on Heun's method, Midpoint method and Ralston's method? 6
(CO1)
(PO1)
3. a) Find the Eigenvalues and Eigenvectors of the following matrix A: 12
(CO2)
(PO1)

$$A = \begin{bmatrix} 2 & 27 & 0 \\ 0 & 4 & 40 \\ 0 & 3 & 30 \end{bmatrix}$$
- b) Describe how linearization transform is applied on the following nonlinear models: 2+2+2
(CO1)
(PO1)
 - i. Exponential model
 - ii. Power model
 - iii. Growth model
- c) Why do we normally use polynomials as interpolating functions? Can single polynomials always be used to accurately interpolate data? 2+5
(CO3)
(PO2)

4. a) For approximating $I = \int_a^b f(x)dx$ using Simpson's 3/8 rule, how are the points chosen over the interval $[a, b]$, and what are their exact coordinates in terms of a and b ? Provide necessary illustrations. (CO1) (PO1) 9
- b) The minimum number of segments for Simpson's 3/8 rule is 3. Given that the 3-segment formula is $I = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$, derive the general n-segment formula? (CO1) (PO2) 8
- c) What is the LU decomposition of A , where $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$? (CO2) (PO1) 8
5. a) What is the motivation behind the use of LU decomposition? Provide a scenario where the use of LU decomposition can enhance the performance of a numerical analysis system. (CO3) (PO2) 5+5
- b) Given n data pairs $(x_1, y_1), \dots, (x_n, y_n)$, do the values of the constants a_0 and a_1 in the least square regression line model $y = a_0 + a_1x$ correspond to the absolute minimum of the sum of the squares of residuals? Are the constants of regression unique? (CO1) (PO2) 9+6
6. a) To maximize the catch of bass in a lake, it is suggested to throw the line to the depth of the thermocline. You can infer that there is a thermocline at a particular depth by noticing the sudden change in temperature. The following table provides us the depth vs temperature data for the lake: (CO2, CO3) (PO3) 6+7+3

Table 1: Depth vs Temperature

Depth, $h(m)$	Temperature, $T(^{\circ}C)$
0	19.1
-1	19.1
-2	19
-3	18.8
-4	18.7
-5	18.3
-6	18.2
-7	17.6
-8	11.7
-9	9.9
-10	9.1

It is noticed that there is a large difference in temperature from $h = -7$ to $h = -8$, so the thermocline must be somewhere between these two depths. The intermediate temperature data between these two depths is missing.

Determine the temperature at $h = -7.5$ using a **first order Lagrange polynomial** and then a **second order Lagrange polynomial** and find the absolute relative approximate error.

- b) Derive Simpson's 1/3 rule using the method of coefficients. Using necessary illustrations, determine the geometric interpretation of this derivation? (CO1) (PO1) 5+4